Critical Site Percolation

A conditional theorem, and the two statements that remain

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Draft, work in progress. Two statements in this note are not yet proved. They are printed in this colour wherever they appear, and they are collected in Section 5. Everything else stated here is proved and machine-checked in Lean, with no unproved step and no axioms beyond the three that Lean itself uses.

Abstract

Declare each site of the integer lattice open with probability p, independently of the others, and ask whether the origin belongs to an infinite open cluster. The probability of that is zero for small p and positive for large p, and the value it takes at the threshold between the two is not known in three or more dimensions. This note reduces that value to two finite statements. The first is a single inequality about connection probabilities, which covers bond percolation and site percolation at once. The second is a reduction of the lattice to a slab of finite width. From these two, and from nothing else, the chance of an infinite cluster at the threshold is zero.

1 The question

In site percolation each site of the lattice is open with probability p and closed with probability $1 - p$, independently of every other site. The open clusters are the connected components of the set of open sites, and $\theta(p)$ is the probability that the origin belongs to an infinite one.

Raising p opens more sites and can only enlarge open clusters, so θ is nondecreasing. There is therefore a critical probability p_c with $\theta(p) = 0$ for $p < p_c$ and $\theta(p) > 0$ for $p > p_c$. Does $\theta(p_c) = 0$?

In two dimensions the answer is yes. For bond percolation, in which edges rather than sites are open, it takes two results together: Harris proved that the percolation probability vanishes at $p = 1/2$, and Kesten proved that $1/2$ is the critical probability. For site percolation it is due to Russo, whose argument follows Kesten's determination of the bond threshold on the square lattice. In three and more dimensions the answer is yes for bond percolation, by a machine-checked proof of Conjecture 3 of Kozma and Nitzan. That settled the dimensions three to ten, the only ones still open, since the lace expansion had already given the result for $d \geq 19$, by Hara and Slade, and then for $d \geq 11$, by Fitzner and van der Hofstad. For site percolation in three or more dimensions it is not known in general, and that is the case treated here.

2 One inequality for both models

It is classical that site percolation is the more general of the two models. Fisher observed that bond percolation on a graph is site percolation on the graph whose vertices are its edges, two of them adjacent when they share an endpoint. The converse fails: Gladkov and Zimin proved that bond percolation cannot simulate site percolation even approximately, already near a vertex of degree three. A statement covering both models must therefore be proved in a structure containing both.

One such structure is the following. Take a finite set of sites and a finite set of labels, and attach to each label some collection of sites. Each label is open with its own probability, independently of the others. Two sites are connected when a chain of open labels joins them, consecutive labels in the chain sharing a site. Connection is reflexive, so every site is connected to itself.

Two familiar models are instances. If every label is attached to exactly two sites, the labels are edges and this is bond percolation. If instead each site carries one label of its own, attached to that site and to one extra point for each of its edges, then a label is open exactly when its site is open, and this is site percolation. The second description is proved below.

Kozma and Nitzan reduced $\theta(p_c) = 0$ to a single inequality about connection probabilities in finite graphs: if that inequality holds in every finite graph, the lattice theorem follows. The first step of the proof is to carry their statement over from graphs to the structure just described. The widening is genuine and not a reformulation: their setting is the case in which every label is attached to exactly two sites, and by the theorem of Gladkov and Zimin the site case cannot be recovered from that one.

Definition 2.1 (the finite inequality)**.** For a nonempty set A of sites and sites o and b,

$$\mathbb{P}(o \leftrightarrow b \text{ and } o \leftrightarrow A) \;\geq\; \mathbb{P}(o \leftrightarrow A)\, \min_{a \in A} \mathbb{P}(a \leftrightarrow b).$$

Read from right to left, it says that reaching the set A and then reaching b from the worst point of A is no better than reaching b directly. What Kozma and Nitzan conjectured is the corresponding statement for bond percolation on a finite graph, which here is the case of labels attached to two sites,

$$\mathbb{P}(o \leftrightarrow b) \;\geq\; \mathbb{P}(o \leftrightarrow A)\, \min_{a \in A} \mathbb{P}(a \leftrightarrow b).$$

Definition 2.1 differs from it in two respects. A label may be attached to any number of sites, which is what reaches site percolation, and the event $\{o \leftrightarrow A\}$ is retained on the right. The second difference makes Definition 2.1 the stronger statement, and the displayed inequality follows from it by discarding that event. For labels attached to two sites the displayed inequality is proved and machine-checked.

Definition 2.1 in the stated generality is not yet proved.

3 Site percolation is a model of labels

The description of site percolation promised above is exact, not an analogy.

Theorem 3.1. *Let a finite graph be given, with each site open with its own probability. Give each site one label, attached to that site and to one further point for each edge at it, and keep that label exactly when the site is open. Then two distinct sites are connected in the model of labels precisely when they are connected by a path of open sites in the graph.*

Proof. Since the labels are the sites, the two models have the same sample space and the same probabilities, so only the connection relation has to be matched. An open label attached to a site reaches, besides that site, one further point for each edge at it, and the only other label reaching that point is the one of the site at the other end of the edge. A chain of open labels from one site to another therefore alternates between sites and the points that stand for the edges between them, and reading off the sites gives a path of open sites. Conversely a path of open sites gives such a chain. □

Corollary 3.2. *The inequality of Definition 2.1 yields the same inequality for site percolation on every finite graph, with no site required to be open in advance.*

The exposure record

The induction behind the inequality exposes the labels incident to a set Z and records what those labels reach outside Z. Write $T(I)$ for the set of sites outside Z reached by a set I of labels.

For graphs, I can be recovered from $T(I)$: a label incident to two sites, one of them in Z, is named by the other. For labels incident to more than two sites it cannot. Two labels incident to the same sites have the same trace, and so do a label reaching $\{a, b\}$ and one reaching $\{b, c\}$ once b has been exposed. The consequence is an asymmetry,

$$T(I \cup J) = T(I) \cup T(J), \qquad T(I \cap J) \subseteq T(I) \cap T(J),$$

with the second inclusion strict in both examples. The induction applies the four functions theorem to $T(I)$, $T(J)$, $T(I \cap J)$ and $T(I \cup J)$, so it needs both relations to be equalities. Recording I itself restores them, since unions and intersections of label sets are unions and intersections. A two-label instance of the strict inclusion is machine-checked.

4 From the inequality to the lattice

The passage from a finite inequality to the lattice is a renormalization: the lattice is divided into blocks, a block is declared occupied when a suitable connection is present inside it, and the occupied blocks are compared with ordinary percolation on a coarser lattice. Carrying that comparison out requires connecting one block to the next, and the classical way to do it is to keep a few extra sites beyond those the construction has already used. Grimmett and Marstrand do exactly this and prove that percolation at a parameter forces percolation in a slab of finite width at any slightly larger parameter. They also record that they cannot remove the excess, and that this is what stops their argument from settling the value at the threshold.

The inequality of Definition 2.1 is what replaces those extra sites. Conditioning on a shell and applying the inequality to the relays it exposes joins one block to the next at the original parameter rather than a larger one, and that step stands exactly where sprinkling used to. What it yields is the following.

Definition 4.1 (the slab reduction)**.** If the chance of an infinite cluster at a parameter strictly inside the unit interval is positive, then percolation occurs in a slab of finite width, at the same parameter or at a smaller one.

Definition 4.1 is not yet proved.

Everything from here to the conclusion is proved. Two routes are available, and they differ in which form of the slab reduction they consume.

Theorem 4.2 (the self-contained route)**.** *Suppose that percolation at a parameter strictly inside the unit interval forces percolation in a slab of finite width at some strictly smaller parameter. Then the chance of an infinite cluster at the threshold is zero.*

Proof. A slab sits inside the lattice, and percolation is monotone under enlarging the vertex set, so percolation in the slab at a parameter gives percolation in the lattice at that parameter. Suppose the chance of an infinite cluster at the threshold were positive. The threshold lies strictly inside the unit interval, so the hypothesis applies and produces a smaller parameter at which the lattice percolates. That contradicts the threshold being the infimum of the parameters at which it does. □

This route assumes nothing else. No theorem about percolation at a critical point enters it, and the two facts that the threshold is neither zero nor one are themselves proved, the first by counting paths and the second by comparing site percolation with bond percolation.

Theorem 4.3 (the route through a half-space)**.** *Suppose that percolation at a parameter forces percolation in a slab of finite width at the same parameter. Suppose in addition that the chance of an infinite cluster in a half-space vanishes at the threshold of the half-space, and that in three or more dimensions this threshold agrees with the one of the lattice. Then the chance of an infinite cluster at the threshold is zero.*

The two additional suppositions are published theorems. The first is due to Barsky, Grimmett and Newman, who prove it for bond percolation and for site percolation alike. The second is due to Grimmett and Marstrand, whose paper works throughout with site percolation. Both are needed, since the first alone places the statement at the threshold of the half-space rather than at the threshold of the lattice.

5 What remains

One conjecture is not yet proved, together with one derivation from it and the standard local facts that derivation consumes.

1. The inequality of Definition 2.1, for labels attached to any number of sites. For labels attached to two sites it is proved.

2. The slab reduction of Definition 4.1.

The two are not on the same footing. The first is a conjecture. The second is not: Kozma and Nitzan derive their slab reduction from their inequality by conditioning on a shell and applying the inequality to the relays that shell exposes, and for bond percolation that derivation is itself machine-checked. What is missing for sites is to carry it out again, which is long but is not open mathematics. The derivation also consumes standard facts about a supercritical parameter in a finite box, local uniqueness and the hitting of each face, which are the finite-volume face of the theorem of Burton and Keane and are not proved here either.

All of these appear in the formalization as named hypotheses, so every statement that uses them displays them and nothing can quietly depend on them. Given them, the conclusion follows by Theorem 4.2 with no further input.

6 The formalization

The definitions, the theorems and the corollary above are formalized in Lean 4 with Mathlib. The model, the connection events, the percolation probability and the critical parameter are all definitions in the files. The critical parameter is proved to lie in the unit interval, so $\theta(p_{\mathrm{c}})$ is defined and the conclusion needs no side condition.

Also proved there, and used above: that conditioning on the whole cluster of a set factorizes exactly; that two disjoint clusters may be factorized in either order; that any probability expands as a finite sum over the possible clusters; that the chance of an infinite cluster is monotone both in the parameter and in the underlying graph; that it vanishes below the threshold; that the threshold is neither zero nor one; that events depending on finitely many sites have probabilities varying Lipschitz continuously in the parameter; and that an exploration whose every step succeeds with probability at least a fixed amount, whatever has happened before, reaches any upward-closed target at least as often as an independent sequence would.

Every one of those compiles with no unproved step, and each is accepted using only the three axioms that Lean itself provides.

References

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